```
use 11

db.students.drop()
db.total_marks_by_city.drop()

db.students.insertMany([
  { _id: 1, name: "Amit", age: 15, marks: 85, city: "Delhi" },
  { _id: 2, name: "Priya", age: 14, marks: 90, city: "Mumbai" },
  { _id: 3, name: "Ravi", age: 16, marks: 78, city: "Kolkata" },
  { _id: 4, name: "Simran", age: 17, marks: 88, city: "Bangalore" },
  { _id: 5, name: "Arjun", age: 15, marks: 92, city: "Delhi" },
  { _id: 6, name: "Meera", age: 16, marks: 81, city: "Mumbai" },
  { _id: 7, name: "Karan", age: 17, marks: 75, city: "Kolkata" },
  { _id: 8, name: "Sonia", age: 14, marks: 95, city: "Bangalore" },
  { _id: 9, name: "Vikram", age: 16, marks: 80, city: "Delhi" },
  { _id: 10, name: "Anjali", age: 15, marks: 89, city: "Mumbai" }
])

var mapFunction = function() {
  emit(this.city, this.marks);
};

var reduceFunction = function(key, values) {
  return Array.sum(values);
};

db.students.mapReduce(
  mapFunction,
  reduceFunction,
  {
    out: "total_marks_by_city"
  }
)

db.total_marks_by_city.find().pretty()

var finalizeFunction = function(key, reducedValue) {
  var output = {
    city: key,
    total_marks: reducedValue
  };
  return output;
};


db.total_marks_by_city.find().pretty()
```

```
// ------------------------------------------------------------
// 🧩 COMMENT SECTION (for report / viva)
// ------------------------------------------------------------
//
// 🔹 DATABASE USED: sagar_11
// 🔹 COLLECTION: students
// 🔹 OUTPUT COLLECTION: total_marks_by_city
//
// ------------------------------------------------------------
// 🔹 MAP FUNCTION:
//   Emits key-value pairs, with each student's city as key and marks as value.
//
// 🔹 REDUCE FUNCTION:
//   Receives city as key and all marks as values array, then returns their sum.
//
// 🔹 FINALIZE FUNCTION:
//   (Optional) Processes the reduced result before output.
//
// ------------------------------------------------------------
// 🔹 EXPECTED OUTPUT SUMMARY:
//   Bangalore → 88
//   Delhi     → 85
//   Kolkata   → 78
//   Mumbai    → 90
//
// ------------------------------------------------------------
// 🔹 VIVA SHORT ANSWERS:
//
// Q1️⃣ : What is MapReduce in MongoDB?
// ➤ MapReduce is a data processing paradigm used to perform
//    large-scale aggregation by mapping documents and reducing results.
//
// Q2️⃣ : What do `map()` and `reduce()` do?
// ➤ `map()` emits key-value pairs.
// ➤ `reduce()` processes those pairs to combine values for each key.
//
// Q3️⃣ : What is the use of `emit()`?
// ➤ It sends a key-value pair from the map function to the reduce phase.
//
// Q4️⃣ : When do we use MapReduce instead of Aggregation?
// ➤ MapReduce is preferred for very large datasets or when custom
//    JavaScript logic is needed during aggregation.
//
// Q5️⃣ : What is the `finalize` function used for?
// ➤ It modifies the result after reduction, e.g., formatting or rounding values.
//
// ------------------------------------------------------------
```

```
// 🔹 CONCLUSION:
// This program demonstrates how MapReduce can be used to process and
// summarize large amounts of data in MongoDB efficiently. It performs
// city-wise total marks calculation using map, reduce, and finalize stages.
// ----------------------------------------------------------
```